Activity 3.6 – Arithmetic Algorithm Units

1.Design Approach:

Designed arithmetic algorithm modules using Verilog HDL to calculate the greatest common divisor (GCD) and product of two 32-bit input values. Implemented iterative subtraction logic for GCD calculation.

2.Verification Steps:

- Created a separate testbench.
- Applied different operand combinations.
- Tested GCD calculation.
- Verified multiplication operation.
- Compared outputs with expected values.
- Simulated using ModelSim.

3.Results:

The testbench checks GCD and multiplication outputs for different input values. PASS/FAIL messages are displayed.

4.Issues Faced:

The iterative algorithm requires proper termination conditions. Zero-valued operands need special consideration. Arithmetic overflow must be considered.

5.Conclusion:

Successfully designed and verified GCD and multiplication algorithm units using Verilog HDL.